

Jai Woods

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Education

University of California, Santa Barbara

Double Major in Economics B.A. and Geography (GIS) B.A.

Santa Barbara, CA
Sept. 2022 - June 2026

Relevant Experience

GIS & Cartography Subcontractor

Barefoot Earthworks (Part-time)

Sonoma County, CA
July 2026 – Present

- Delivering comprehensive geospatial site analysis, topographic mapping, and grading layouts for a 100+ acre retreat property development.
- Developed high-resolution DEMs/DSMs, slope profiles, and watershed drainage models in QGIS to guide heavy equipment earthwork and access routing.
- Produced topographic maps, satellite hybrid views, and hill aspects to streamline municipal permitting reviews and contractor site execution.

Esri Student Assistant

Esri

San Diego, CA
July 2026

- Selected for the competitive Esri Student Assistantship program to support technical sessions, user workshops, and event operations at the annual Esri User Conference in San Diego, supporting the core operational roles at the store, wristband, and registration desks.
- Expanded knowledge of Utility Network data models, digital twins, and ArcGIS Pro workflows for utility applications through technical sessions within the Esri ecosystem, as well as innovations in geospatial technology implementation.

UAS Technician and Geospatial Data Researcher

University of Iowa Physics and Astronomy Department

Iowa City, IA
July 2025 – August 2025

- Engineered custom 3D-printed payload housing to secure LiDAR and RGB sensor arrays onto a DJI Matrice 300 RTK drone.
- Integrated microcontrollers, Raspberry Pi, battery power systems, and sensors via custom circuitry; developed automated Python pipelines to collect, structure, and integrate raw sensor and geospatial data.
- Conducted 6 UAS data acquisition flights capturing LiDAR point clouds and photogrammetric data over preserved grassland sites.
- Executed spatial suitability and canopy height modeling using LiDAR datasets and Structure from Motion (SfM) photogrammetry; performed geospatial data quality control (QC) and presented findings at a university research symposium.

Apprentice Tradesman

Part-time

Sonoma County, CA
May 2019 – June 2026

- Performed residential electrical, construction, plumbing, landscaping, and property-maintenance work while in school, working under various contractors.
- Planned and executed residential electrical and plumbing layouts, operated heavy equipment, and adapted work to field conditions.
- Managed project design, budgeting, scheduling, and quality control while maintaining a 100% commitment to job-site safety and risk mitigation.

Technical Research Intern

The Nature Conservancy (Bren Environmental Leadership program)

Santa Barbara, CA
June 2024 – September 2024

- Authored comprehensive technical workflow documentation and deployment guides for deep-learning based whale detection software to establish standardized use documentation for users.
- Collaborated with NOAA and The Nature Conservancy scientists to integrate technical documentation across software platforms, supporting data integrity and consistency across the software user guide.
- Applied convolutional neural networks (CNNs) and machine-learning methodology frameworks to execute spatial image classification and large-scale geospatial queries.

UCSB Geography Club Co-President

UCSB Geography Department

Santa Barbara, CA
Sept. 2025 - June 2026

- Participated in organizing club events, student engagement, administrative tasks and planning throughout the school year.

House Treasurer

Isla Vista Student Housing Co-op

Santa Barbara, CA
Sept. 2025 - Dec. 2025

- Audited \$21,000+ in total monthly rent and utilities payments for accounting errors, discrepancies, and late payments.
- Coordinated with Housing Co-op board members to resolve transaction issues, manage budgeting, and ensure organized bookkeeping.

Projects

Pumped Hydroelectric Suitability Modeling

Apr. – June 2025

- Performed multi-criteria spatial suitability and risk analysis for pumped hydroelectric energy storage in California using ArcGIS Pro and geodatabase layers.
- Integrated datasets for high-voltage transmission lines, land use, slope, solar potential, existing water bodies and zoning; evaluated constraints and documented results selecting 5 ideal sites for pumped hydroelectric storage.

Gacho Real Estate Association Underwriting Project

May 2026

- Underwrote a \$1.9M, 8-unit investment property in Buellton, CA. Built a comprehensive pro forma and presentation outlining a five-year business plan and exit strategy for the mock-up investment deal. Presented our project proposal to real estate investors for feedback.

Skills and Certificates

GIS & Esri Products: 3+ years using GIS software, including ArcGIS Pro, ArcGIS Online, QGIS, ENVI, Google Earth Engine

Geospatial Data & Databases: Geodatabases, spatial data integration, database management, quality control (QC), LiDAR processing, remote sensing, image classification, network analysis, suitability and risk analysis

FAA Part 107 Licensed Drone Pilot: Photography, videography, photogrammetry, mapping, 3D modeling, LiDAR collection and inspections

Programming & Querying: Python (pandas, GeoPandas), SQL, R, C, C++, C#, JavaScript, HTML, CSS

Technical Support & Communication: Technical-session and GIS-workshop support, workflow documentation, troubleshooting, and cross-functional collaboration

Microsoft Office & Google Suite: Excel, Word, PowerPoint; Docs, Sheets, Slides, Forms, Drive

Quantitative Coursework: Econometrics, economics, finance, accounting, financial statements, pro formas, and financial modeling

Permaculture Design Certificate (Permaculture Artisans)